

Brice Peres

PhD candidate – Computer Graphics & Physical Simulation

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Portfolio: briceprs.github.io ↗



Experience

Inria Elan team – PhD candidate

Numerical study of naturally-curved fibre assemblies

Apr. 2026 – present

Grenoble, France

- Numerical modelling and simulation of bundles of naturally-curved fibres (continuation of the work below).

Inria Elan team – Pre-PhD researcher

Tomographic hair-data filtering

Sept. 2025 – Mar. 2026

Grenoble, France

- Designed filtering methods for tomographic acquisitions of human hair, in preparation of the PhD project.

Dassault Systèmes – End-of-studies intern

3D engine performance optimisation

Sept. 2024 – Mar. 2025

Paris, France

- Worked with the rendering team to improve the performance of one of Dassault Systèmes' main 3D engines.

Corys – R&D intern

Unreal Engine 5, runtime train simulator

May 2023 – Sept. 2023

Grenoble, France

- Implemented real-time windscreen effects (rain, snow, dirt, cleaning) on Unreal Engine 5 (HLSL).
- Worked in the R&D team led by Alain Kocelniack.

Inria – Research intern

GPU terrain filtering

June – July 2022

Grenoble, France

- Designed a GPU height-map filtering algorithm for terrain generalisation (advisors: Joëlle Thollot, Romain Vergne).

Selected projects

Real-time soft-body simulation

C++, CUDA

Spring-mass soft body accelerated on the GPU with CUDA.

Dec. 2024

Progressive raytracer

C++, OpenGL

GPU raytracer with BVH traversal, shadows, reflections, refractions and environment maps. Cook-Torrance shading with a GGX microfacet distribution; frame accumulation for progressive refinement.

Dec. 2024

Particle physics simulation

C++

2D/3D simulation under gravity and a Lennard-Jones potential with custom spatial data structures. Ran 8 000 particles over 400 000 steps in 20 minutes on a laptop.

May 2023

Black Seas – multiplayer naval game

Rust, OpenGL – team of 5

From-scratch engine; rendering, terrain, lighting, post-processing (ray-marched clouds, bloom, tonemapping).

Feb. – Apr. 2023

Education

ENSIMAG – Engineering degree (M.Eng.), Grenoble INP

Computer science and applied mathematics

2021 – 2024

Grenoble, France

Institute of Science Tokyo (formerly Tokyo Tech)

Academic exchange semester

Mar. – Aug. 2024

Tokyo, Japan

Lycée Notre-Dame de Sion – Classes préparatoires MPSI/MP

Two-year intensive course for French engineering school exams

2018 – 2021

Marseille, France

Skills, languages & interests

Languages. C++, Rust, C, Python, GLSL, HLSL

Graphics & GPU. OpenGL, CUDA, WebGL, Unreal Engine 5, Shadertoy

Tools. Git, $\LaTeX$, Linux • **Familiar.** Java, Shell, Assembly

Spoken. French (native) • English (C1, TOEFL 100 pts) • Japanese (A1)

Interests. Climbing, bouldering & hiking since 2011 (a few climbing competitions); computer graphics, physically based rendering, real-time simulation, Shadertoy.